13.1 Model 129

Note:

- Universal ATA control modules (used on models 124, 129, 140, 202) must be version coded according to menu point 5 on the HHT display.

- ATA can be activated and deactivated using the mechanical key.

Control module manufacturers:

Becker, Temic

Diagnosis – Technical Changes

Version start dates/Changes/Innovations

World wide Manuf. code	Model	LHS[1] RHS[1]	Manuf. plant	As of chassis number	Up to chassis number	As of production date	Up to production date	Type and reason for change	Reference/Remarks
WDB	129					01/94		Vehicles using IRCL (code 880) can not be activated or deactivated via mechanical key.	
WDB	129					06/97		Introduction of DAS2	
WDB	129					06/97		Activation and deactivated of ATA via mechanical key and lock nut switch.	(only USA)
WDB	129					06/97		CH version deleted.	
WDB	129					06/98		Alarm siren (H3) is being replaced with alarm siren with auxiliary battery (H3/1) in model production. Anti-tow sensor included with optional version of ATA.	

1) LHS: Left hand steering
RHS: Right hand steering

Diagnosis – Function Test

⚠
After performing the Function Test, erase any stored DTC’s (see section 0).

Test step/Test scope	Test condition	Nominal value	Possible cause/Remedy [1]
⇒ 1.0 ATA status indication	• Lock vehicle, using transmitter key. • Wait approx. 15 seconds	ATA status indicator (E33) **or** LED in ATA status/towing protection switch (S85/3)	23 ⇒ 19.0 23 ⇒ 20.0
⇒ 2.0 Activate ATA via left door	• Open driver’s window. • Lock vehicle with transmitter key. • After approximately 15 seconds open door from inside.	Alarm horn sounds, lights flash. (then deactivate ATA)	23 ⇒ 6.0
⇒ 3.0 Activate ATA via right door	• Open passenger window. • Lock vehicle with transmitter key. • After approximately 15 seconds open door from inside.	Alarm horn sounds, lights flash. (then deactivate ATA)	23 ⇒ 6.0
⇒ 4.0 Activate ATA via engine hood	• Open driver’s window. • Lock vehicle with transmitter key. • Wait approximately 15 seconds. • Release engine hood through open window. • Open engine hood.	Alarm horn sounds, lights flash. (then deactivate ATA)	23 ⇒ 8.0

[1] Observe Preparation for Test, see 22.

Diagnosis – Function Test

Test step/Test scope	Test condition	Nominal value	Possible cause/Remedy [1]
⇒ 5.0 Activate ATA via trunk lid	• Open trunk lid. • Lock vehicle with transmitter key. • Turn off trunk lamp (rotary trunk lid microswitch [S88/1] open). • Wait approximately 15 seconds. • Turn on trunk lamp (rotary trunk lid microswitch [S88/1] closed).	Alarm horn sounds, lights flash. (then deactivate ATA)	23 ⇒ 7.0
⇒ 6.0 Activate Anti-tow protection	• Lock vehicle with transmitter key. • Wait approximately 15 seconds. • Lift vehicle using provided vehicle jack (lift vehicle only at lift points), until wheel clears ground.	Up to 05/97 As of 06/97 Alarm horn sounds, lights flash. (then deactivate ATA)	23 ⇒ 22.0 23 ⇒ 23.0
⇒ 7.0 Deactivate Anti-tow protection	• Ignition: **OFF** • Press ATA status/towing protection switch (S85/3).	ATA status/towing protection switch (S85/3) illuminates for approx. 2 seconds.	23 ⇒ 24.0 23 ⇒ 20.0
⇒ 8.0 *Not applicable for U.S.A. vehicles*			
⇒ 9.0 *Not applicable for U.S.A. vehicles*			
⇒ 10.0 *Not applicable for U.S.A. vehicles*			

1) Observe Preparation for Test, see 22.

Diagnosis – Diagnostic Trouble Code (DTC) Memory

Preparation for Test
1. Fuses OK.
2. Battery voltage 11 – 14 V
3. Ignition: **ON**
4. Connect HHT, see section 0

Electrical wiring diagrams:
Electrical Troubleshooting Manual, Model 129.

DTC	Possible cause	Hints	Test step/Remedy [1]
No communication with HHT	Diagnostic line	Check diagnostic output line.	23⇒ 14.0
001	No DTC fault stored in memory	If complaint exists, perform entire Electrical Test Section.	23
002	Alarm triggered via rotary tumbler/trunk lid microswitch (S88/1)		If false alarm, see 23⇒ 7.0
003	Alarm triggered via engine hood switch (ATA) (S62)		If false alarm, see 23⇒ 8.0
004	*Not applicable for U.S.A. vehicles*		
006	Alarm triggered via left/right front door switch (S17/3, S17/4)		If false alarm, see 23⇒ 6.0
007	Alarm triggered via ATA tow sensor (B33)	Up to 05/97 As of 06/97	If false alarm, see 23⇒ 22.0 23⇒ 23.0

[1] Observe Preparation for Test, see 22.

Diagnosis – Diagnostic Trouble Code (DTC) Memory

DTC	Possible cause	Hints	Test step/Remedy [1]
010	Alarm triggered via radio (A2)	Applies to older generation of radio only	If false alarm, see 23⇒ 10.0
012	Alarm triggered via ignition/starter switch (S2/1)		If false alarm, see 23⇒ 11.0
014	Alarm triggered via stop lamp switch (4-pole) (S9/1)	Up to 05/97	If false alarm, see 23⇒ 9.0
019	ATA control module (N26)		23⇒ 1.0
020	Ground contact missing for left door actuator (S47)		Left door not locked properly, S47
021 Control module: **Temic**	Short circuit after circuit 30 while in armed status, results in starter lock-up.	From 10/93 to 02/94 only.	23⇒ 18.0
Becker	Blinker system has short circuit after circuit 30	Not valid for U. S. A. vehicles.	
022 Control module: **Temic**	Short circuit after circuit 30 while in armed status.		23⇒ 1.0
Becker	Short circuit after circuit 30 while in armed status, results in starter lock-up.	From 10/93 to 02/94 only.	23⇒ 18.0

[1] Observe Preparation for Test, see 22.

Diagnosis – Diagnostic Trouble Code (DTC) Memory

DTC	Possible cause	Hints	Test step/Remedy [1]
022 Control module: **Becker**	Open circuit in circuit 30 while in armed status.		23⇒ 1.0
026	Alarm triggered via alarm siren with auxiliary battery (H3/1)		If false alarm, see 23⇒ 13.0
027	Alarm triggered via alarm siren with auxiliary battery (H3/1), short circuit after ground		23⇒ 13.0
028	Alarm triggered via alarm siren with auxiliary battery (H3/1), communication interrupted		23⇒ 13.0

[1] Observe Preparation for Test, see 22.

Diagnosis – Complaint Related Diagnostic Chart

Preparation for Test:
1. Review 10, 21, 22, 31,
2. Fuses OK,
3. Battery voltage 11 – 14 V
4. IRCL system functional, if not see Test Section 4.10 in DM, B&A, Vol. 2.
5. CL system functional, if not see Test Section 3.5 in DM, B&A, Vol. 1.
6. Ignition: ON

Electrical wiring diagrams:
Electrical Troubleshooting Manual, Model 129

Complaint/Problem	Possible cause	Test step/Remedy [1]
Entire ATA system does not function.	ATA control module (N26)	23 ⇒ 1.0
ATA system does not function when using IRCL.	Connection IRCL - ATA disturbed: With IRCL up to 05/97 as of 06/97	 23 ⇒ 2.0 23 ⇒ 3.0
ATA system does not function when using mechanical key.	Mechanical lock switch: Up to 05/97 With IRCL up to 12/93	23 ⇒ 4.0
No confirmation signal when arming the ATA system.	Doors have not been completely closed, ATA status indicator (E33) (w/o anti-tow sensor) **or** ATA status/towing protection switch (S85/3) (with anti-tow sensor)	 23 ⇒ 19.0 23 ⇒ 20.0
ATA status indicator does not function.	ATA status indicator (E33) (w/o anti-tow sensor) **or** ATA status/towing protection switch (S85/3) (with anti-tow sensor)	23 ⇒ 19.0 23 ⇒ 20.0

[1] Observe Preparation for Test, see 22.

Diagnosis – Complaint Related Diagnostic Chart

Complaint/Problem	Possible cause		Test step/Remedy [1]
ATA does not sound alarm when opening a front door.	Left front door switch (S17/3) Right front door switch (S17/4)		23 ⇒ 6.0
ATA does not sound alarm when opening the trunk lid.	Rotary tumbler/trunk lid microswitch (S88/1)		23 ⇒ 7.0
ATA does not sound alarm when opening engine hood.	Engine hood switch (ATA) (S62)		23 ⇒ 8.0
ATA does not sound alarm when removing the radio.	ATA contact connector (radio) (1-pole) (applies to older radio generation up to 05/97)		23 ⇒ 10.0
ATA "acoustical" alarm does not sound.	Alarm horn (H3) **or** Alarm siren with auxiliary battery (H3/1).	Up to 05/98 As to 06/98	23 ⇒ 12.0 23 ⇒ 13.0
ATA "optical" alarm does not function.	Front headlamps (only USA) Taillamps (only USA)		23 ⇒ 16.0 23 ⇒ 17.0
ATA will not cancel via remote control.	IRCL or RCL (as of 06/97)		Check system, see DM, B&A, Vol. 2, Test Section 4.10
ATA does not sound alarm via anti-tow function.	Anti-tow control module (N26) ATA tow sensor (B33)		23 ⇒ 22.0 23 ⇒ 23.0

[1] Observe Preparation for Test, see 22.

Electrical Test Program – Component Locations

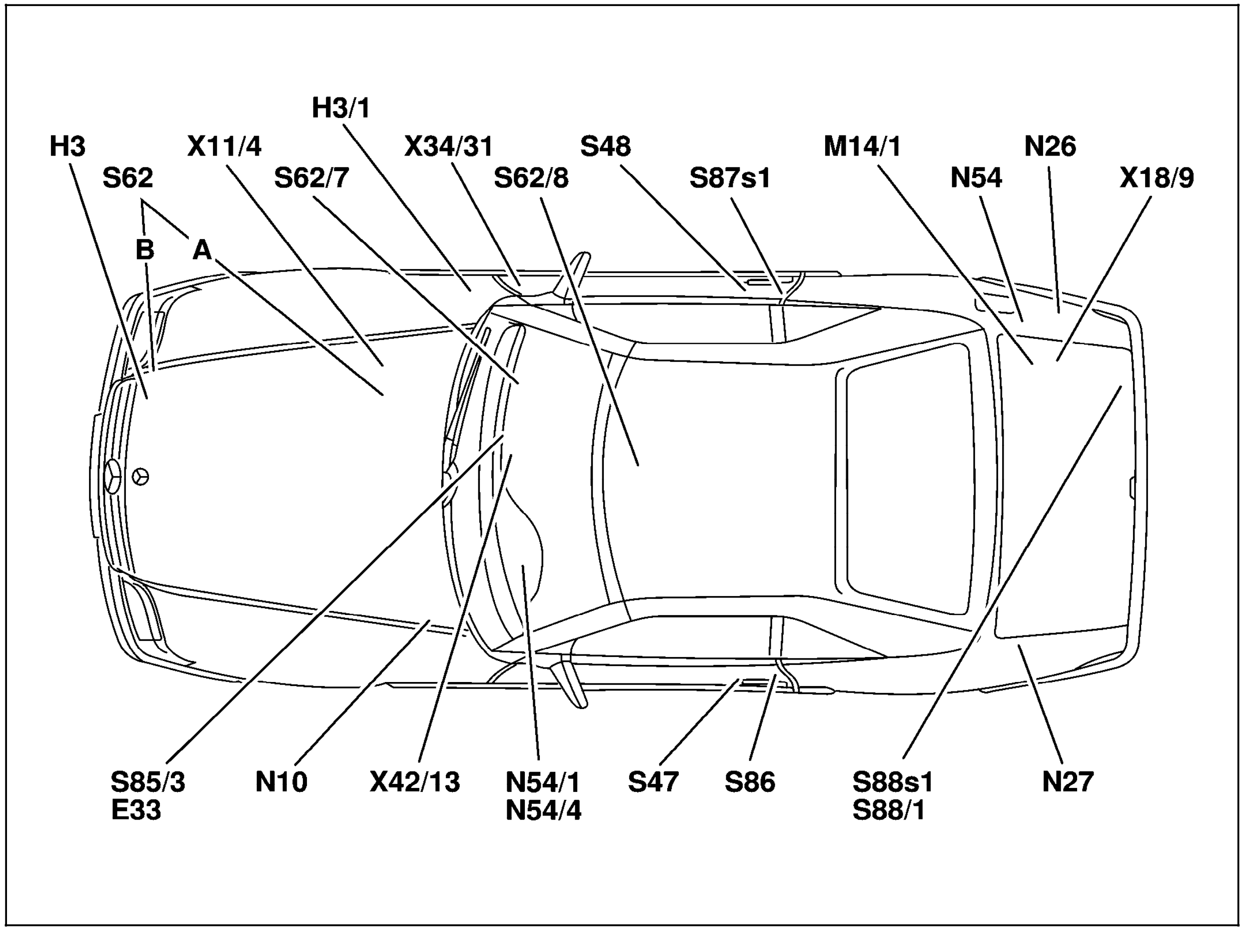

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Figure 1

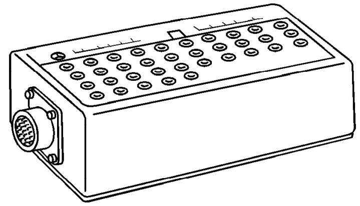
E33	ATA status indicator
H3	Alarm horn (up to 05/98)
H3/1	Alarm horn with auxiliary battery (as of 06/98)
N26	ATA control module
N27	Anti-tow control module (not USA), repl. with B33
N54	IRCL control module
N54/1	DAS control module
N54/4	DAS radio frequency/infrared control module (as of 06/97)
S47	Left door actuator
S48	Right front door actuator
S62	Engine hood switch
S62/7	Glove box switch (ATA) (not USA)
S62/8	Center console compartment switch
S85/3	ATA status/towing protection switch
S86	Left door switch group
S87s1	ATA/CF microswitch
S88s1	ATA/CF microswitch
S88/1	Rotary tumbler/trunk lid microswitch

Electrical Test Program – Preparation for Test

1. Review 10, 21, 22, 31,
2. Fuses are OK,
3. Battery voltage 11–14 V,
4. IRCL system functional, if not see Test Section 4.10 in DM, B&A, Vol. 2.
5. CL system functional, if not see Test Section 3.5 in DM, B&A, Vol. 1.
6. Entire exterior lighting system (blinkers, head and tail lamps) is functional.

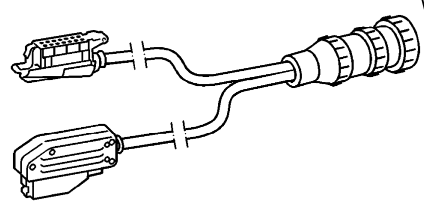
Electrical wiring diagrams:
Electrical Troubleshooting Manual, Model 129

Special Tools



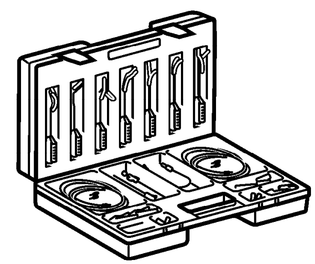
124 589 00 21 00

35-pin socket box



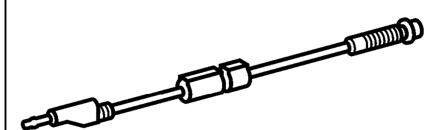
129 589 04 63 00

33-pin test cable



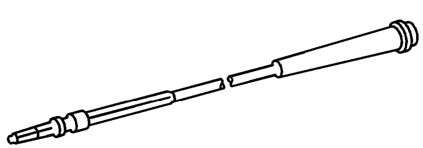
201 589 00 99 00

Electrical connecting set



124 589 37 63 00

Fused cable



140 589 22 63 00

Adapter cable

Electrical Test Program – Preparation for Test

Test equipment; See MBUSA Standard Service Equipment Program

Description	Brand, model, etc.
Digital multimeter	Fluke models 23, 77 III, 83, 85, 87

Electrical Test Program – Preparation for Test

Connection Diagram – Socket Box
(as shown on model 129)

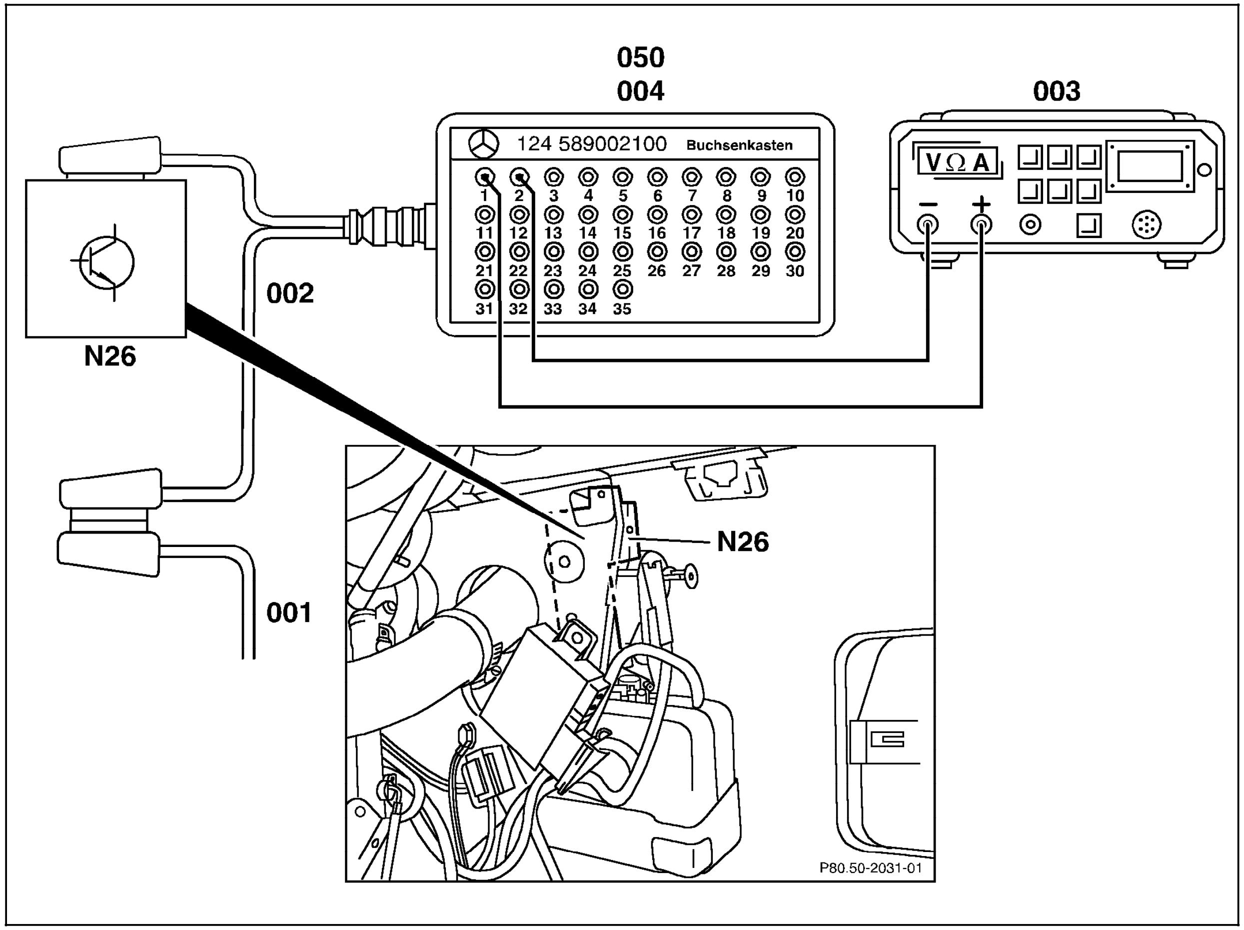


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Figure 1

001	ATA control module connector
002	Test cable
003	Digital multimeter
004/050	Socket box
N26	ATA control module

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
1.0	019	**ATA control module (N26)** Voltage supply Circuit 31	N26 4 — V — +		11 – 14 V	Wiring.
1.1		Circuit 30	N26 4 — V — 1		11 – 14 V	Wiring.
2.0		**Signal from CL Open/Close** with IRCL up to 05/97	N26 11 — V — 1	Unlock vehicle via IRCL	11 – 14 V	Wiring, IRCL control module.
3.0		**Lock nut switch signal from CL Open** Actual value (as of 06/97)		HHT actual value, Locknut switch 1 Unlock vehicle using IRCL or mechanical key (only USA)	no YES	⇒ 3.2
3.1		Lock nut switch signal from CL Close Actual value		HHT actual value, Locknut switch 2 Unlock vehicle using IRCL or mechanical key (only USA)	no YES	⇒ 3.3

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
3.2		Opening Signal Voltage	N26 11 — –V+ — 1	Unlock vehicle using IRCL or mechanical key (only USA)	11 – 14 V	Wiring, IRCL/RCL, see DM, B&A, Vol. 2, Test Section 4.10
3.2		Closing Signal Voltage	N26 19 — –V+ — 1	Unlock vehicle using IRCL or mechanical key (only USA)	11 – 14 V	Wiring, IRCL/RCL, see DM, B&A, Vol. 2, Test Section 4.10
4.0		**Left ATA/CF microswitch (S86)** (up to 05/97) with IRCL (up to 12/93)	N26 11 — –V+ — 1	Insert mechanical key into left door lock, turn in direction lock or unlock and hold.	11 – 14 V	Wiring, S86
4.1		Right ATA/CF microswitch (S87s1) (up to 05/97) with IRCL (up to 12/93)	N26 11 — –V+ — 1	Insert mechanical key into right door lock, turn in direction lock or unlock and hold.	11 – 14 V	Wiring, S87s1
4.2		Rotary tumbler/trunk lid microswitch (S88s1)	N26 11 — –V+ — 1	Insert mechanical key into trunk lid lock, turn in direction lock or unlock and hold.	11 – 14 V	Wiring, S88s1

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
5.0	004	***Not applicable to U.S.A. vehicles***				
6.0	006	**Door contact switch** Actual values		HHT actual values. Door contact switch LF/RF: Open left door Close left door Open right door Close right door	 ON OFF ON OFF	⇒ 6.1
6.1		Left front door switch (S17/3) Contact	N26 24 — V — 1	Open left door	11 – 14 V	Wiring, S17/3
6.2		Right front door switch (S17/4) Contact	N26 24 — V — 1	Open right door	11 – 14 V	Wiring, S17/4
7.0	002	**Rotary tumbler/trunk lid microswitch (S88/1)** Actual values		HHT actual values Rotary tumbler/trunk lid microswitch (S88/1) lamp: ON (lamp illuminates) S88/1 OFF	 ON OFF	⇒ 7.1

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
7.1		Contact	N26 26 — V — 1	S88/1 ON S88/1 OFF	11 – 14 V < 1 V	Wiring, S88/1
8.0	003	**Engine hood switch (S62)** Actual values		Engine hood closed: Engine hood open:	ON OFF	⇒ 8.1
8.1		Contact	N26 27 — V — 1	Engine hood closed: Engine hood open:	< 1 V 11 – 14 V	Wiring, S62, N26
9.0	002	**Stop lamp switch (S9/1)** Actual values (up to 05/97)		Ignition: ON Service brake not pressed. Press service brake:	OFF ON	⇒ 9.1
9.1		Contact	N26 4 — V — 23	Ignition: ON Press service brake	< 1 V 11 – 14 V	Wiring, S9/1

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
10.0	010	**Radio contact** Actual values (only older radio generation up to 05/97)		HHT actual values Radio contact, vehicle with radio Vehicle without radio or contact not made.	 On OFF	⇒ 10.1
10.1		Contact	N26 17 — 1	Vehicle with radio Vehicle witout radio or contact not made.	11 – 14 V < 1 V	Wiring, radio contact (X42/13)
11.0	012	**Ignition alarm circuit** (up to 05/97)	N26 4 — 18	Ignition: OFF Ignition: ON	< 1 V 11 – 14 V	Wiring, Ignition/starter switch (S2/1)
12.0		**Alarm siren (H3)** Activation (up to 05/98)		ON	Alarm sounds.	⇒ 12.1
12.1		Function	N26 17 — 13	**Use bridges with 124 589 37 63 00 safety cables only.**	Alarm sounds.	Wiring, H3

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
13.0		**Alarm siren with auxiliary battery (H3/1)** Function (as of 06/98)		HHT activation: Alarm siren test	Alarm siren sounds short acoustical signal (0.2 sec)	⇒ 13.1
13.1		Voltage supply	H3/1 2 —(◄−V+►)— 1	Disconnect connector at H3/1	11 – 14 V	Wiring.
13.2		Trigger alarm		Activate ATA. Wait 15 seconds. Disconnect connector at H3/1. To stop alarm tone, reconnect connector at H3/1 and deactivate alarm.	There is both a acoustical and optical alarm actuation.	H3/1 Wiring, N26
14.0		**Diagnostic Output** Contact	N26 X11/4 22 —(◄−Ω+►)— 11	Disconnect connector on N26	11 – 14 V	Wiring.
15.0		*Not applicable to U.S.A. vehicles*				
15.1		*Not applicable to U.S.A. vehicles*				

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
16.0	002	**Alarm circuit** **Headlamps** Activation (only USA)		HHT activation: Headlamps (only USA)	Headlamps: ON	⇒ 16.1
16.1		Function	N26 1 — 12	**Use bridges with 124 589 37 63 00 safety cables only.**	Headlamps: ON	Wiring, Headlamp unit, n26
17.0		**Alarm circuit** **Taillamps** Activation (only USA)		HHT activation: Taillamps	Left/rightTail/ parking lamp (E3e2, E4e2) illumin. steady.	⇒ 17.1
17.1		Left taillamp circuit Function	N26 1 — 14	**Use bridges with 124 589 37 63 00 safety cables only.**	Tail/parking lamp (E3e2) illuminates steady.	Wiring, E3e2
17.2		Right taillamp circuit Function	N26 1 — 3	**Use bridges with 124 589 37 63 00 safety cables only.**	Tail/parking lamp (E4e2) illuminates steady.	Wiring, E4e2, If values are OK: N26

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
18.0		**Starter lock-out circuit (K38)** Activation (from 10/92 to 12/93 only)		HHT activation: Starter lock-out ON, Ignition/starter switch is in position 3 (circuit 50)	Engine does not start.	⇒ 18.1
18.1		Resistance	N26 5 — Ω — 18	Igntion: **OFF**	50 – 60 Ω	Wiring, K38
18.2		Circuit 15	N26 5 — V — 18	Ignition: **ON**	11 – 14 V	Wiring, Circuit 15
18.3		Ground	N26 ⊥ — V — 4	Ignition: **ON**	11 – 14 V	Wiring.
18.4		Circuit 31 from ATA control module (N26)	K38 5 — V — 4	Ignition: **ON**	11 – 14 V	Wiring, N26

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
18.5		Starter lock-out relay module (K38) Circuit 50	K38 3 — −V+ — 1	Igntion: **OFF**	11 – 14 V	Wiring, K38, Ignition/starter switch (S2/1), Starter.
19.0		**ATA status indicator (E33)** Activation (without anti-tow protection)		HHT activation	E33 illuminates.	⇒ 19.1
19.1		Function	N26 8 — — 1	**Use bridges with 124 589 37 63 00 safety cables only.**	E33 illuminates.	Wiring, E33, N26
20.0		**ATA status/towing protection switch (S85/3)** Activation of the status indication (with anti-tow protection)		HHT activation: ATA status indicator LED: ON	LED in switch illuminates.	⇒ 20.1
20.1		Function	N26 8 — — 1	**Use bridges with 124 589 37 63 00 safety cables only.**	LED in switch illuminates.	Wiring, S85/3, N26

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
21.0	002	**Model recognition** Actual values (up to 05/97)		HHT actual values: Model	 129	⇒ 19.1
21.1		Ground activation	N26 33 — V — 1	i Socket 33 must be contacted.	11 – 14 V	Wiring.
22.0		**Ant-tow control module (N27)** Activation (up to 05/97)		HHT activation: Anti–tow ON Wait 15 seconds. Lift vehicle at lift point using vehicle jack.	ON	⇒ 22.1
22.1		Voltage supply	N27 5 — V — 2	Activate ATA, Wait 60 seconds, Lift vehicle at lift point using vehicle jack.	11 – 14 V	Wiring, N26 If values are OK: ⇒ 22.2
22.2		Trigger signal	N26 29 — V — 2	Activate ATA, Wait 60 seconds, Lift vehicle at lift point using vehicle jack.	11 – 14 V, only briefly.	Wiring, N27

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
23.0		**ATA tow sensor (B33)** Activation (as of 06/97)		HHT actual values: Anti–tow ON Wait 15seconds. Lift vehicle at lift point using vehicle jack.	ON	⇒ 23.1
23.1		Voltage supply	B33 5 — – V + — 1	Activate ATA	11 – 14 V	Wiring.
23.2		Simulate ATA triggering	B33 1 — — 5	Disconnect connector at B33, Activate ATA, Install bridge, **Use bridges with 124 589 37 63 00 safety cables only.**	Alarm is activated.	Wiring, N26, If values are OK: B33

Electrical Test Program – Test

⇒		Test scope	Test connection	Test condition	Nominal value	Possible cause/Remedy
24.0		**ATA status/towing protection switch (S85/3)** Actual values (with anti-tow protection)		HHT actual values: ATA status/towing protection switch (S85/3) button **not pressed:** **S85/3 button pressed:**	 OFF ON	⇒ 24.1
24.1		Function	N26 8 — 1	**Use bridges with 124 589 37 63 00 safety cables only.**	LED in switch illuminates.	Wiring, S85/3, If values are OK: N26

Control Module – Programming

Menu item 5 on the HHT allows programming of those ATA control modules which require programming. Only upon completion of the programming is the control module operable. The programming is menu driven.

Follow the instructions shown on the HHT display screen.

Coding possibilities	Selections	Hints
Vehicle version	Model 129 up to 05/97 Model 129 as of 06/97 with DAS2	
ATA country version	USA Delayed headlamp shutoff duration, 0 - 120 secs. Belgium Rest of the world	USA: via HHT
Anti-tow protection	Yes No	
Alarm siren (signal) (up to 05/98) B NL GB	Yes No	See sub-menu in HHT for add. countries